# Assignment 1

Fabiano Blasch, Nik Gabrovsek, Patrick Gyasi, Gregor Steiner

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## Time Series Analysis

### 1

We can use the **quantmod** package to download the data from the FRED database.

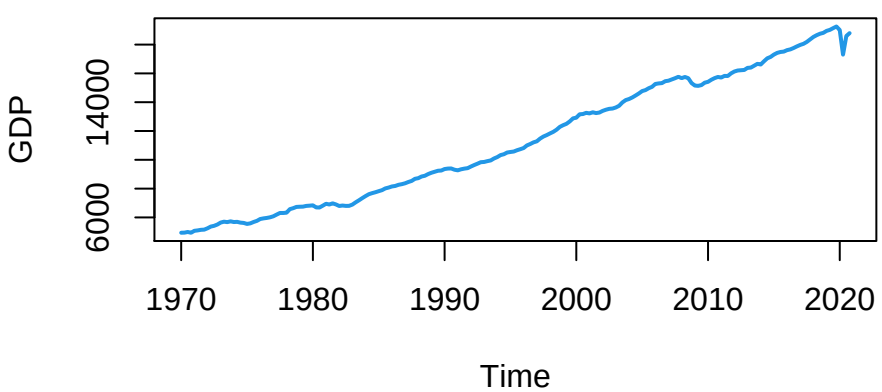
```
# Download data
quantmod::getSymbols("GDPC1", src = "FRED")
```

```
## [1] "GDPC1"
```

```
gdp.ts <- ts(GDPC1, start = c(1947, 1), frequency = 4)

# Filter
gdp.ts <- window(gdp.ts, start = c(1970, 1), end = c(2020, 4))
```

After filtering for the time period we are interested in, we can now plot the series.

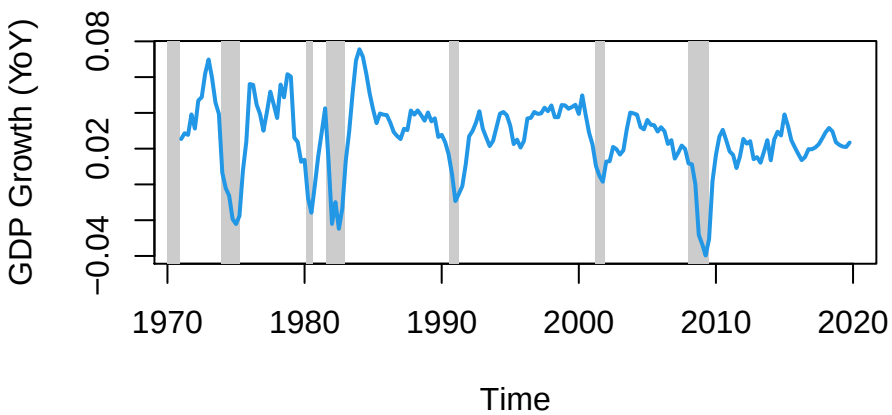
```
# Plot
plot(gdp.ts, lwd = 2, col = 4, ylab = "GDP")
```




### 3

```
# Compute y/y growth rates
gdp.growth <- diff(gdp.ts, 4) / lag(gdp.ts, 4)
```

```r
# Plot
plot(gdp.growth, type = "n", ylab = "GDP Growth (YoY)")
tis::nberShade()
lines(gdp.growth, lwd = 2, col = 4)
```




### 4

```r
# Test for stationarity
tseries::adf.test(gdp.growth)
```

```
## Warning in tseries::adf.test(gdp.growth): p-value smaller than printed p-value
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  gdp.growth
## Dickey-Fuller = -4.5746, Lag order = 5, p-value = 0.01
## alternative hypothesis: stationary
```

We do reject the null-hypothesis of non-stationarity. Thus, the series is stationary.

### 5

We want to estimate the model

$$y_t = c + \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \varepsilon_t$$

in a Bayesian fashion. In matrix notation this reads as

$$y = X\varphi + \varepsilon,$$

where $y := [y_3, ..., y_T]', \varphi := [c, \varphi_1, \varphi_2]', \varepsilon := [\varepsilon_3, ..., \varepsilon_T]'$ and

$$X := \begin{bmatrix} 1 & y_2 & y_1 \\ \vdots & \vdots & \vdots \\ 1 & y_{T-1} & y_{T-2} \end{bmatrix}.$$

We obviously lose the first two observations, leaving us with $T - 2$ observations.

The first step is choosing prior distributions for $c, \varphi_1, \varphi_2$. A simple normal prior might be a good start. Thus,

$$\varphi \sim N_3(\mu_0, \Sigma_0),$$

where $\mu_0 = [\mu_c, \mu_{\varphi_1}, \mu_{\varphi_2}]'$ and $\Sigma_0 = diag(\sigma_c^2, \sigma_{\varphi_1}^2, \sigma_{\varphi_2}^2)$.

Then, we only need a prior distribution for the error variance, where we will use an Inverse Gamma Distribution as in the linear model:

$$\sigma^2 \sim IG(c_0, d_0).$$

And we know that

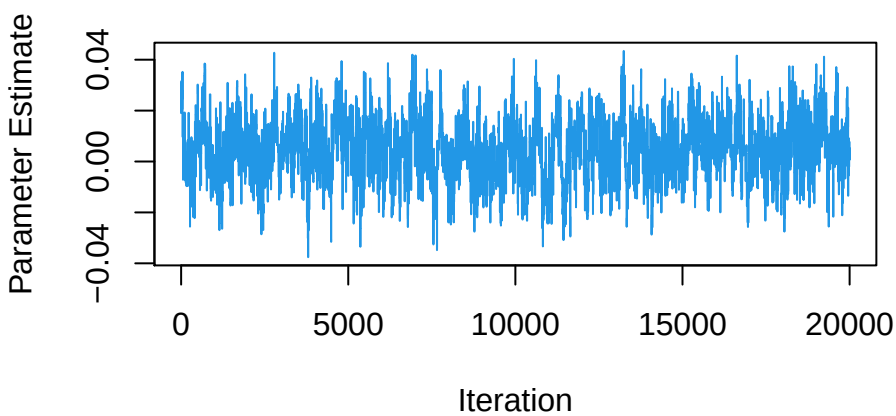
$$y|\varphi, \sigma^2 \sim N_{T-2}(X\varphi, \sigma^2 I_{T-2}).$$

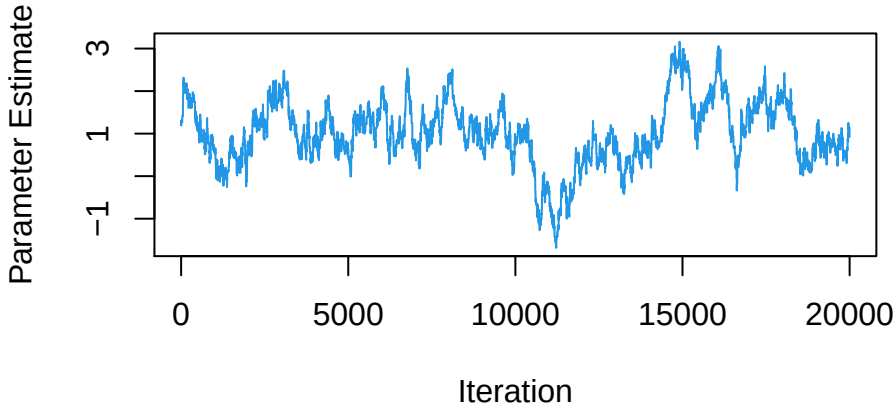
Now, we can start setting up the Metropolis algorithm to estimate the posterior density.

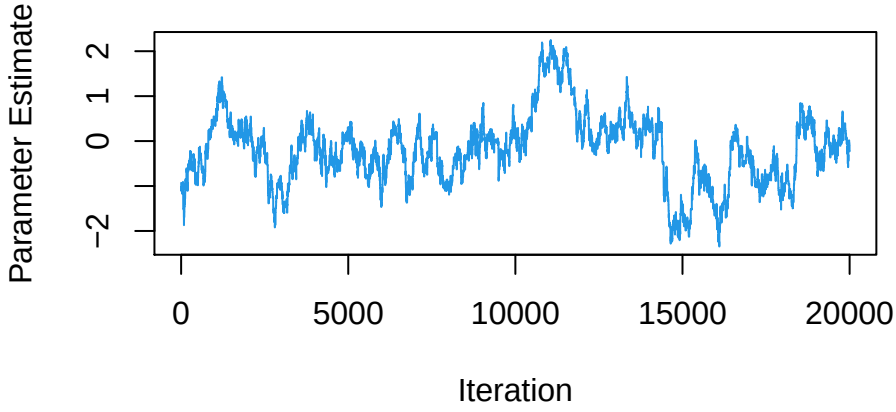
```
########## Metropolis Algorithm ##########


# Set up X matrix and y vector
T <- length(gdp.growth)
X <- cbind(rep(1, T-2), gdp.growth[-c(1, T)], gdp.growth[-c(T-1, T)])
y <- gdp.growth[-c(1, 2)]


# Prior Setup

# Normal prior on phi vector
mu0 <- c(0, 0, 0)
sig2.0 <- 100 # Same variance for the three parameters

# IG prior on sigma2
c0 <- 2
d0 <- 1

# Algorithm Setup
nburn <- 2e4
nsave <- 2e4
ntot <- nburn + nsave
t <- nrow(X)

# proposal draw variances
eta2.c <- 0.05
eta2.phi1 <- 0.4
eta2.phi2 <- 0.4
eta2 <- c(eta2.c, eta2.phi1, eta2.phi2)

# starting values
sig2.draw <- 1
phi.draw  <- c(0, 0, 0)
accept    <- 0


# Storage matrix
phi.store <- matrix(NA, nsave, 3)
colnames(phi.store) <- c("c", "phi1", "phi2")
```

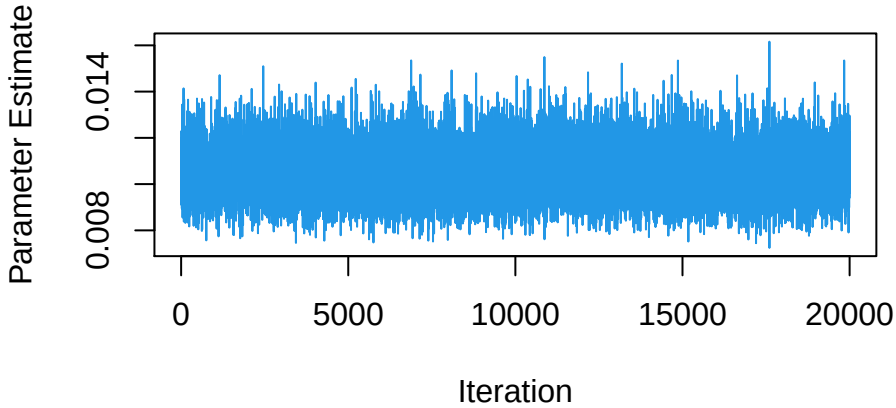
```r
sig.store <- matrix(NA, nsave, 1)


# Metropolis algorithm loop
for (i in 1:ntot) {

  ## STEP 1:

  # Create proposal from normal distribution
  #phi.prop <- MASS::mvrnorm(1, phi.draw, diag(sqrt(eta2)))
  phi.prop <- mapply(function(x, y) rnorm(1, x, sd = sqrt(y)), phi.draw, eta2)

  # Evaluate prior for proposal
  prior.current <- sum(dnorm(phi.draw, mu0, sqrt(sig2.0), log = TRUE))
  prior.prop    <- sum(dnorm(phi.prop, mu0, sqrt(sig2.0), log = TRUE))

  # Evaluate LL
  ll.current <- sum(dnorm(y, X %*% phi.draw, sqrt(sig2.draw), log = TRUE))
  ll.prop <- sum(dnorm(y, X %*% phi.prop, sqrt(sig2.draw), log = TRUE))

  # And add prior
  c.p.current <- ll.current + prior.current
  c.p.prop <- ll.prop + prior.prop

  # And LR
  ll.ratio <- c.p.prop - c.p.current

  if(ll.ratio > log(runif(1))){ #if our proposal is likely "enough"

      #replace the current draw for phi with our proposal
      phi.draw <- phi.prop

      #add 1 to our acceptance counter
      accept   <- accept + 1

  }

  # Tune proposal variance during first 50% of burn-in
  if (i < 0.5 * nburn) {
    if(accept / i > 0.3) eta2 <- eta2 * 1.01
    if(accept / i < 0.15) eta2 <- eta2 * 0.99
  }

  ## STEP 2: SAMPLE SIGMA2 GIVEN PHI & DATA

  # compute errors
  e  <- y - X %*% phi.draw

  # sum of squared residuals (equivalent to sum((e)^2))
  ee <- t(e) %*% e

  # compute posterior quantities
  ck <- c0 +  t/2
```

```
  dk <- d0 + ee/2
  # sample sigma2 from inverse gamma posterior
  sig2.draw <- 1 / rgamma(1, ck, dk)

  # STEP 3: STORE POSTERIOR SAMPLES

  #only after burn-in period!
  if(i > nburn){

  phi.store[i - nburn,]  <- phi.draw
  sig.store[i - nburn,] <- sig2.draw

  }


}


# Plot traceplots
apply(cbind(phi.store, sig.store), 2, function(x) plot.ts(x , xlab = "Iteration", ylab= "Parameter Estim
```



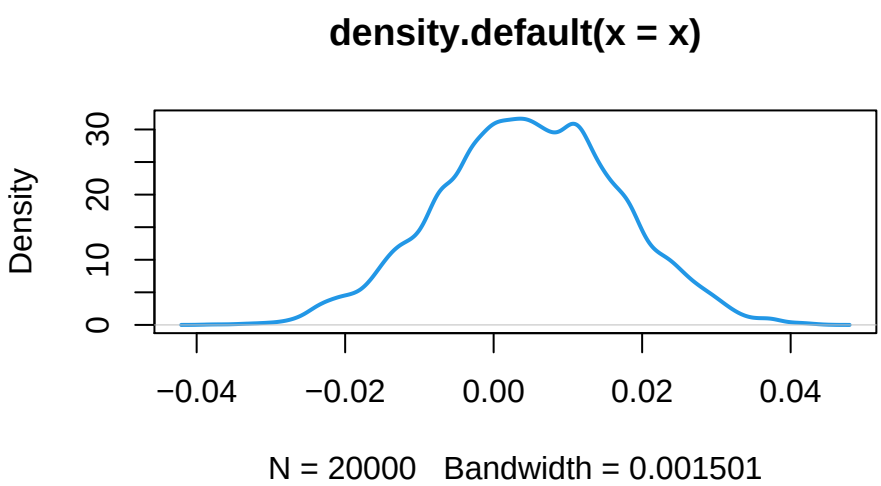

Parameter Estimate
3
1
−1
0
5000
10000
15000
20000
Iteration

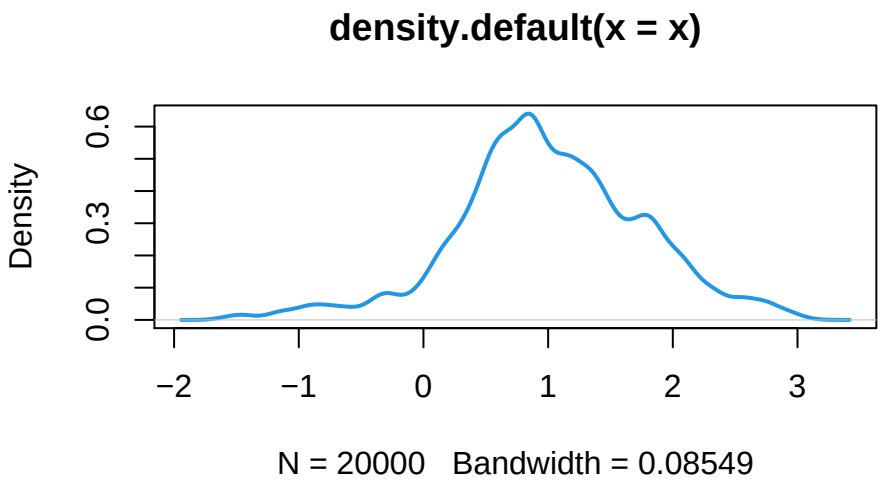

Parameter Estimate
2
1
0
−2
0
5000
10000
15000
20000
Iteration

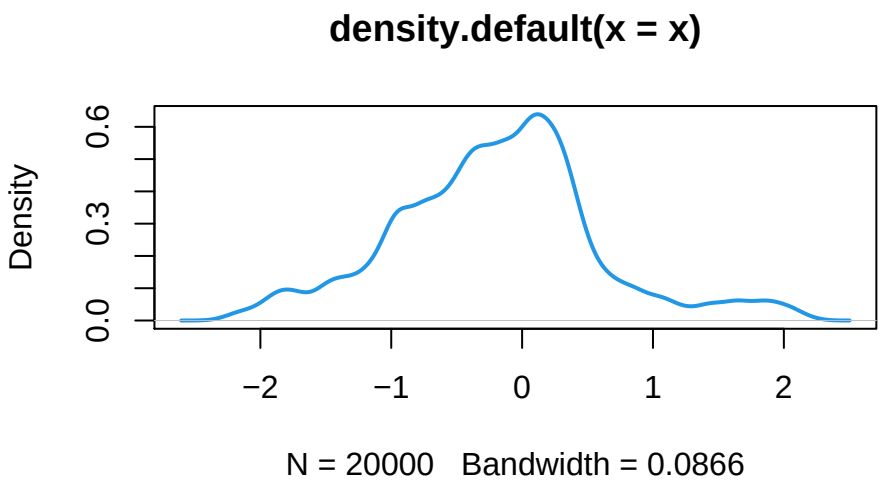


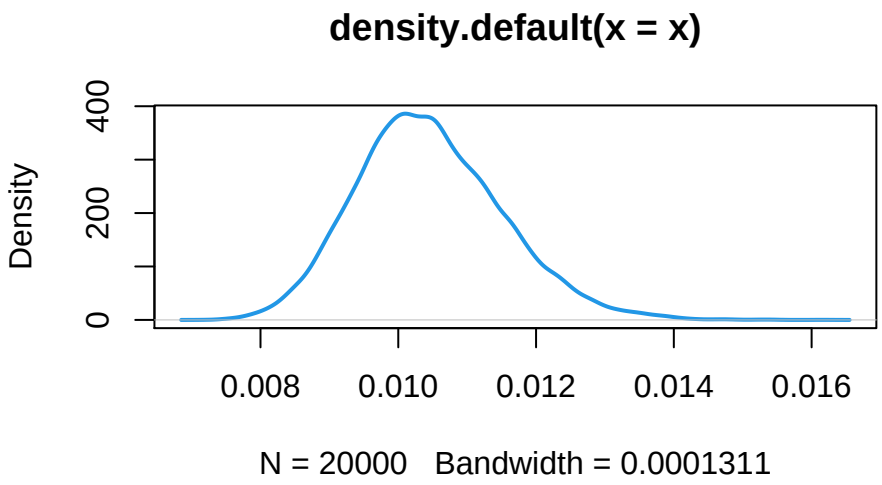

```
## NULL
```

```r
# Plot densities
apply(cbind(phi.store, sig.store), 2, function(x) plot(density(x), lwd = 2, col = 4))
```




density.default(x = x)
Density
0.0
0.3
0.6
−2
−1
0
1
2
3
N = 20000 Bandwidth = 0.08549


density.default(x = x)
Density
0.0
0.3
0.6
−2
−1
0
1
2
N = 20000 Bandwidth = 0.0866




```
## NULL
```

Let's have a look at the posterior distribution the algorithm ended up with.

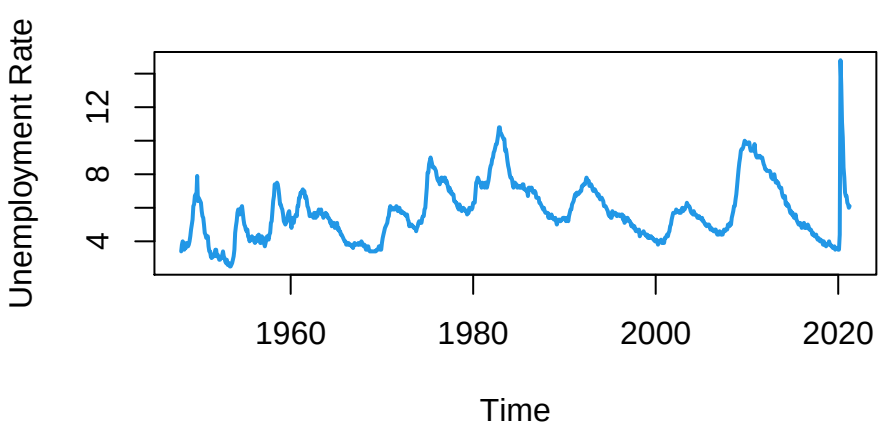
## Prior Sensitivity

Again, we can use the **quantmod** package to download the data from the FRED database.

```r
# Download data
quantmod::getSymbols("UNRATE", src = "FRED")
```

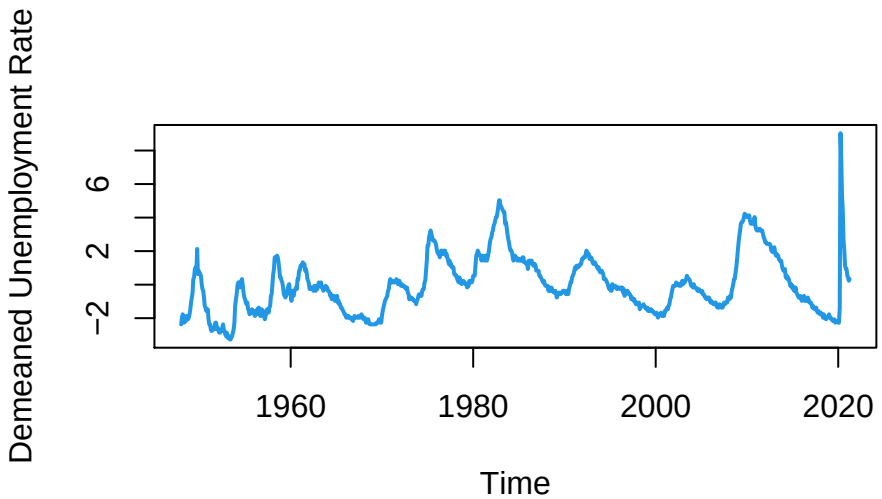
```
## [1] "UNRATE"
```

```r
un.ts <- ts(UNRATE, start = c(1948, 1), frequency = 12)

# Plot
plot(un.ts, col = 4, lwd = 2, ylab = "Unemployment Rate")
```



## 1

```r
# Demean series
un.demeaned <- un.ts - mean(un.ts)

# Plot
plot(un.demeaned, col = 4, lwd = 2, ylab = "Demeaned Unemployment Rate")
```



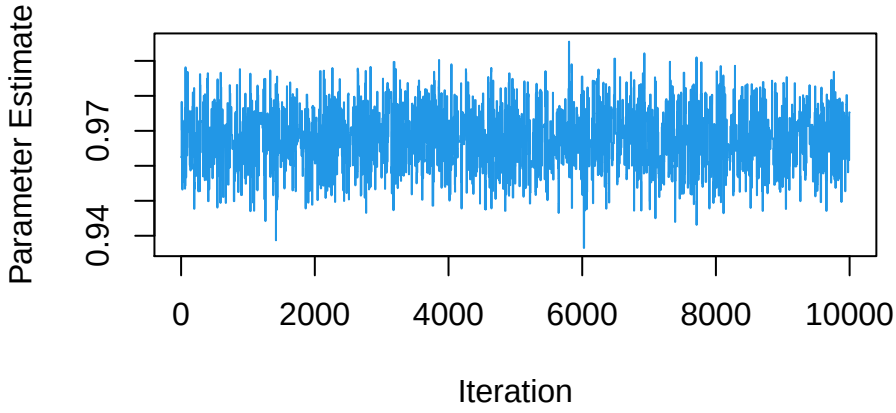

## 2

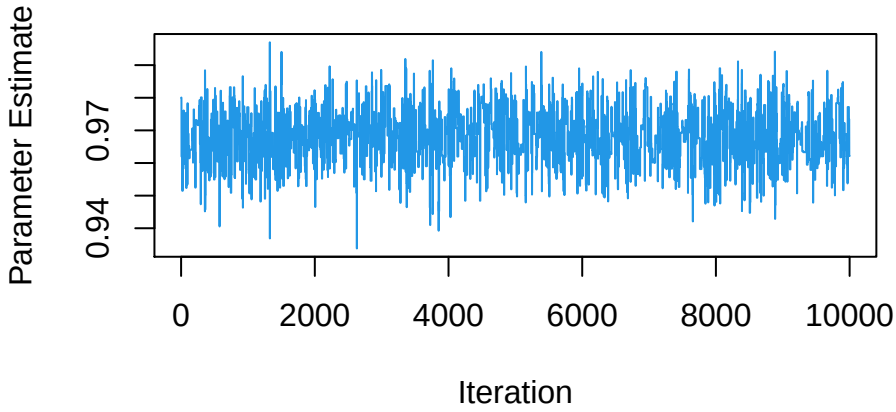
We will use the Metropolis algorithm used in the lecture.

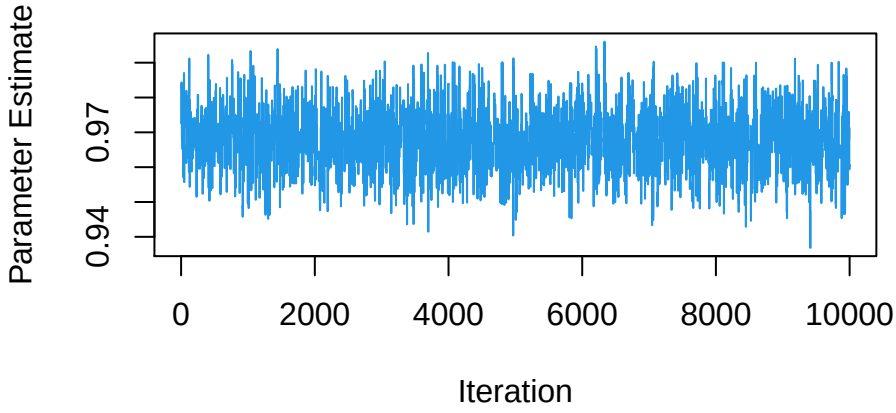
```r
metro.est <- function(y, x, prior, prior.params){


  # IG prior on sigma2
  c0 <- 2
  d0 <- 1

  # Algorithm Setup
  nburn <- 10000
  nsave <- 10000
  ntot <- nburn + nsave
  t <- length(x)
  eta2  <- 0.0161

  sig2.draw <- 1
  phi.draw  <- 0.1
  accept    <- 0


  # Storage matrix
  phi.store <- matrix(NA, nsave, 1)

  sig.store <- rep(NA, nsave)
```

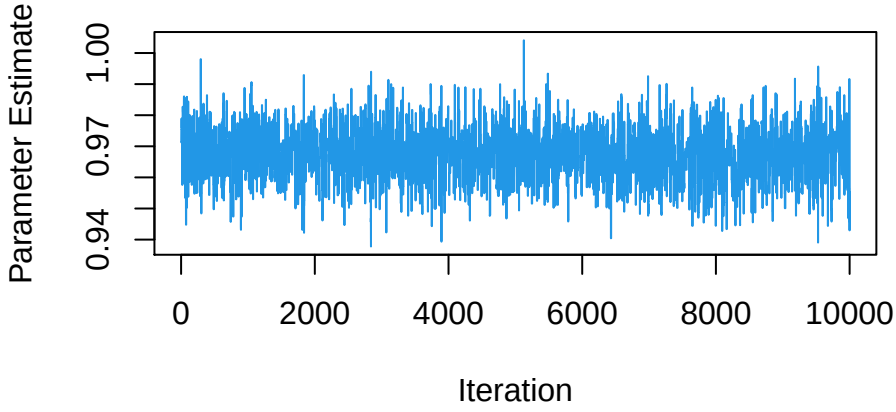
```r
# Metropolis algorithm loop
for (i in 1:ntot) {

  ## STEP 1:

  # Create proposal
  phi.prop <- rnorm(1, phi.draw, sqrt(eta2))


  # Evaluate prior for proposal
  prior.current <- do.call(paste0("d", prior), list(phi.draw, prior.params[1], prior.params[2]))
  prior.prop    <- do.call(paste0("d", prior), list(phi.prop, prior.params[1], prior.params[2]))

  # Evaluate LL
  ll.current <- sum(dnorm(y, x * phi.draw, sqrt(sig2.draw), log = TRUE))
  ll.prop <- sum(dnorm(y, x * phi.prop, sqrt(sig2.draw), log = TRUE))

  # And add prior
  c.p.current <- ll.current + prior.current
  c.p.prop <- ll.prop + prior.prop

  # And LR
  ll.ratio <- c.p.prop - c.p.current

  if(ll.ratio > log(runif(1))){ #if our proposal is likely "enough"

      #replace the current draw for rho with our proposal
      phi.draw <- phi.prop

      #add 1 to our acceptance counter
      accept   <- accept + 1

    }

  # Tune proposal variance during first 50% of burn-in
  if (i < 0.5 * nburn) {
    if(accept / i > 0.3) eta2 <- eta2 * 1.01
    if(accept / i < 0.15) eta2 <- eta2 * 0.99
  }


  ## STEP 2: SAMPLE SIGMA2 GIVEN PHI & DATA

  # compute errors
  e  <- y - x * phi.draw

  # sum of squared residuals (equivalent to sum((e)^2))
  ee <- t(e) %*% e

  # compute posterior quantities
  ck <- c0 +  t/2
  dk <- d0 + ee/2
  # sample sigma2 from inverse gamma posterior
```

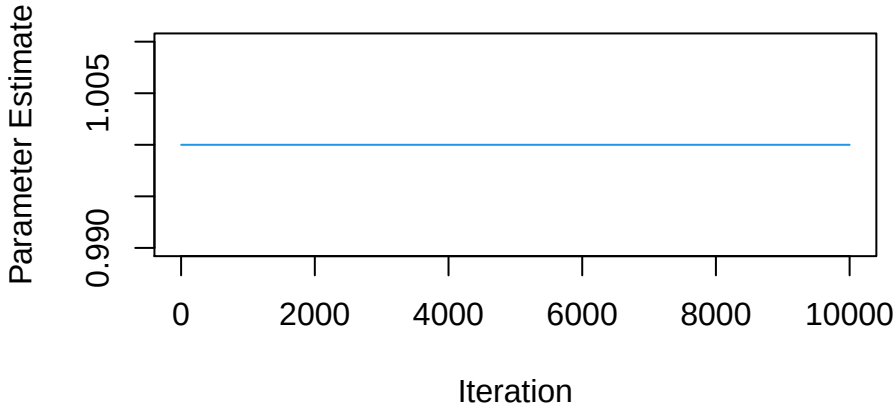
```r
    sig2.draw <- 1 / rgamma(1, ck, dk)

    # STEP 3: STORE POSTERIOR SAMPLES

    #only after burn-in period!
    if(i > nburn){

    phi.store[i - nburn,]  <- phi.draw
    sig.store[i - nburn] <- sig2.draw

    }


  }

  return(list(Phi = phi.store, Sig2 = sig.store))

}

# Estimate posteriors
yy <- un.demeaned[-1]
xx <- un.demeaned[-length(un.demeaned)]

priors <- c("norm", "norm", "beta", "beta", "beta")
prior.params <- list(c(0,1), c(1, 1), c(1, 1), c(10, 10), c(0.1, 0.1))

res <- list()

for (i in 1:length(priors)) {
  res[[i]] <- metro.est(yy, xx, priors[i], prior.params[[i]])
}


# Traceplots and densities
lapply(res, function(x) plot.ts(x[[1]], xlab = "Iteration", ylab= "Parameter Estimate", col = 4))
```


Parameter Estimate
0.94
0.97
0
2000
4000
6000
8000
10000
Iteration


Parameter Estimate
0.94
0.97
0
2000
4000
6000
8000
10000
Iteration

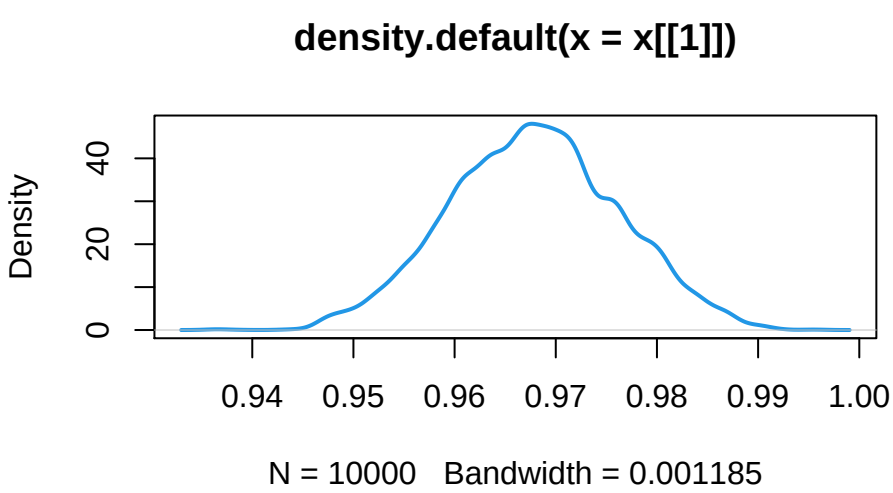

Parameter Estimate
0.94
0.97
0
2000
4000
6000
8000
10000
Iteration

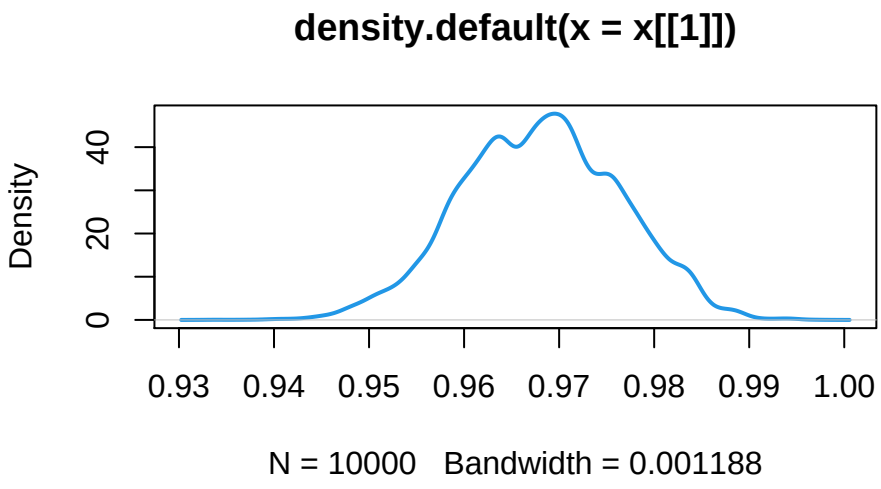

Parameter Estimate
0.94
0.97
1.00
0
2000
4000
6000
8000
10000
Iteration

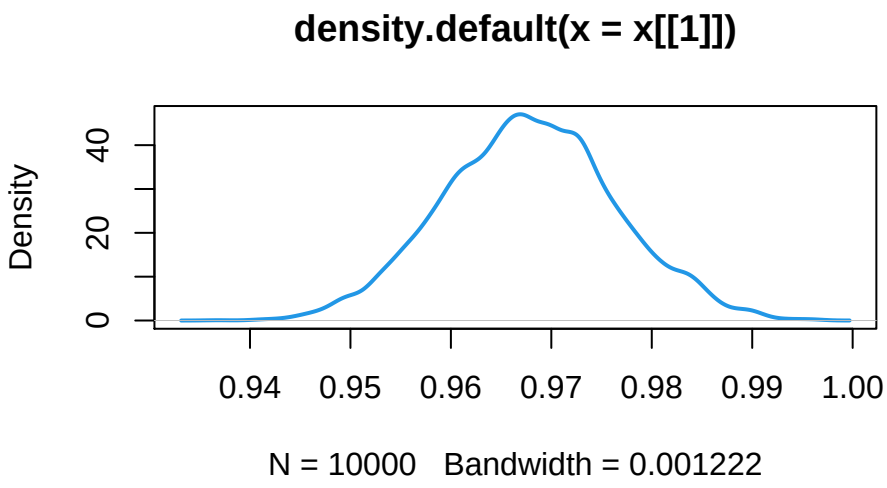



```
## [[1]]
## NULL
##
## [[2]]
## NULL
##
## [[3]]
## NULL
##
## [[4]]
## NULL
##
## [[5]]
## NULL
```

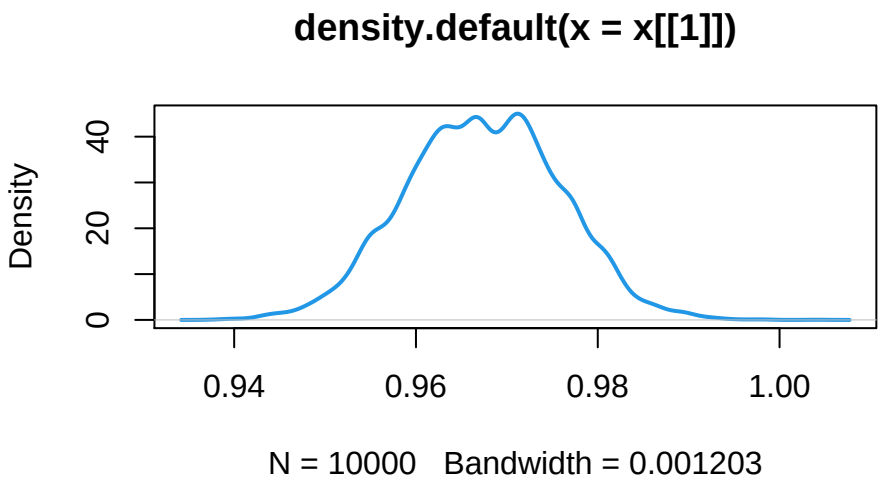
```
lapply(res, function(x) plot(density(x[[1]]), lwd = 2, col = 4))
```



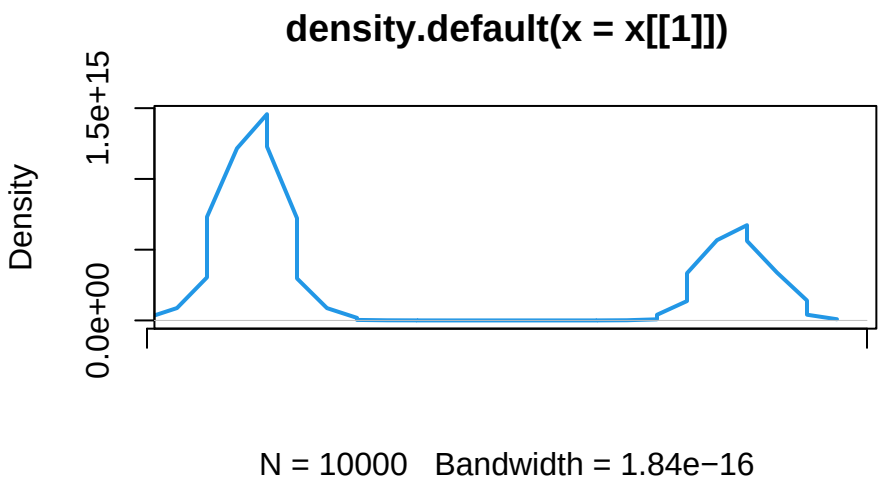



```
## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
## collapsing to unique 'x' values
```




```
## Warning in plot.window(...): relative range of values ( 0 * EPS) is small (axis
## 1)
```




```
## [[1]]
## NULL
##
## [[2]]
## NULL
##
## [[3]]
## NULL
##
## [[4]]
## NULL
##
## [[5]]
## NULL
```